

VMS Phase 1 FSD — Business Partner & Vehicle Creation

2026-09-20 · @Someone

1. Document control

This FSD specifies two modules of VMS Phase 1 in build-ready detail: **Business Partner creation** and **Vehicle creation**. Everything else in the Phase 1 requirement document depends on these two masters, so they are specified first.

Item	Detail
Document	VMS Phase 1 FSD — Business Partner & Vehicle Creation
Version	1.0 — draft for client review
Date	20 Sep 2026
Source requirement	Vehicle Management System — Phase 1 / V1, Updated Functional & Implementation Requirements
Source sections covered	2 (Business Partner), 3 (Categories), 5 Basic + Ownership rows, 8 (Bank lease), 13 (Audit), 14 (Entities)
Audience	Client sponsor, product owner, development team, QA, UAT testers
Currency	PKR, 2 decimal places
Locale	Taken from the machine the system is opened on — its time zone, calendar and date format

In scope

- Business Partner master: create, edit, multi-role assignment, contacts, addresses, bank details, documents, status lifecycle, duplicate control.

- Vehicle master: create with full technical identity, ownership category and its counterparty, acquisition cost, bank lease agreement and schedule, attached items, documents, opening financial postings.
- Field-level definitions, validations, business rules, permissions, audit and acceptance criteria for both.

Out of scope for this document

- Trips, monthly P&L screens, driver mobile app, maintenance jobs, notification delivery engine, reports and dashboards. These are separate FSDs; this document defines only the data and hooks they consume.
- Live GPS, tracker APIs, geofencing, route replay — Phase 2 per the source document.

2. Scope and objectives

The objective is that a user can create a Business Partner once and a Vehicle once, and every downstream module — expenses, income, installments, trips, maintenance, notifications — finds the master data it needs without re-entry.

Four design objectives govern every rule in this document:

1. **One party, many roles.** A workshop that also supplies parts is one record with two roles, not two records.
2. **Capture the financial opening position at creation.** Purchase price, down payment and lease agreement are entered on the vehicle creation screen and posted as system transactions, so the vehicle's paid-to-date figure is correct from day one without any manual total.
3. **Nothing is overwritten.** Category changes, role changes and counterparty changes are closed and superseded, never edited in place.
4. **Minimum typing.** User, timestamp, branch, sequence numbers and derived amounts are supplied by the system.

Dependency map

```

flowchart LR
    BP[Business Partner] --> VEH[Vehicle]
    BP --> FIN[Finance Agreement]
    VEH --> FIN
    VEH --> TXN[Vehicle Transactions]
    FIN --> TXN
    VEH --> DOC[Documents]
    DOC --> NOTIF[Notification Rules]
    FIN --> NOTIF
    VEH --> OPS[Trips / Maintenance]
    BP --> OPS
  
```

Business Partner must be built and populated before Vehicle: a vehicle cannot be created in the Bank Leased, Rented, Shared or Customer Arrangement category without an existing partner to point at.

3. Glossary and conventions

Term	Meaning in this system
Business Partner (BP)	Any external or internal party the company deals with. One master record, one or more roles.
Role	A capacity in which a BP is used: Driver, Workshop, Bank, Vendor, Customer, Tracker Company, Body Maker, Fuel Card Company, Running Customer.
Ownership Category	The commercial relationship of a vehicle: Self Owned, Shared, Rented, Customer Arrangement, Bank Leased.
Counterparty	The BP tied to the vehicle by its current ownership category (the bank, lessor, partner or customer).
Vehicle Relation	A dated record joining a vehicle to a BP in a category. Closed and superseded on change, never edited.
Attached Item	A major removable asset fitted to the vehicle: container, AC unit, tyre set, crane, tank.

Term	Meaning in this system
Opening Posting	A vehicle transaction the system creates automatically at vehicle creation.
Finance Agreement	The lease or financing contract with a bank, holding schedule and installment terms.

Requirement ID scheme

Prefix	Meaning	Example
FR-BP / FR-VH	Functional requirement	FR-VH-014
BR-BP / BR-VH	Business rule, testable	BR-BP-007
VAL-BP / VAL-VH	Field or form validation with a message	VAL-VH-003
AC-BP / AC-VH	Acceptance criterion	AC-BP-002
OQ	Open question for the client	OQ-05

Field table legend

Every field specification table uses these columns. **M** = mandatory, **O** = optional, **C** = conditional (mandatory only when the stated condition is true). **Sys** = system-supplied, never typed by the user.

Data types are written in SQL Server terms: `nvarchar(n)`, `decimal(18,2)` for money, `date`, `datetime2`, `bit`, `int`, `uniqueidentifier`.

Module A — Business Partner

4. Concept and rationale

A Business Partner is one master record for a person or company, carrying one or more roles. Roles are additive flags with their own attribute sets, not separate records and not a single "type" field.

The alternative — a separate Drivers table, Vendors table, Banks table — breaks down the moment the same party appears twice. These cases are real in this business:

Situation	Single-master result	Separate-tables result
A workshop also sells tyres and spare parts	One BP, roles Workshop + Vendor. One ledger, one payables position.	Two records, split history, reconciliation by hand.
An adda pays monthly hire and also owns a vehicle the company runs	One BP, roles Customer + Running Customer + Shared counterparty.	Three records for one party.
A driver leaves and returns as a contractor with his own vehicle	Same BP, Driver role reactivated, new Rented counterparty relation.	New record, old trip and expense history orphaned.
A bank finances vehicles and is also a corporate customer	One BP, roles Bank + Customer.	Two records, confusing selection lists.

Rules that follow from the concept

- **FR-BP-001** — A BP may hold any combination of the nine roles at once. There is no restriction on combinations.
- **FR-BP-002** — Role-specific fields are captured only when that role is active. Adding the Driver role reveals licence fields; removing it hides them but retains the stored values.
- **FR-BP-003** — Every selection list elsewhere in the system filters by role, not by name. The bank picker on a vehicle shows only BPs with an active Bank role.
- **FR-BP-004** — A BP is either a **Person** or a **Company**. This choice drives which identity fields apply (CNIC vs NTN/STRN) and cannot be changed after the first transaction is posted against the BP.

5. Business Partner data model

Nine tables. `BusinessPartner` is the master; everything else hangs off it.

flowchart TD

```

BP[BusinessPartner] --> ROLE[BusinessPartnerRole]
BP --> CON[BPContact]
BP --> ADDR[BPAddress]
BP --> BANK[BPBankAccount]
BP --> DOC[BPDocument]
ROLE --> DRV[BPDriverDetail]
ROLE --> VEN[BPVendorDetail]
ROLE --> CUS[BPCustomerDetail]

```

Entity	Holds	Cardinality to BP	Key columns
BusinessPartner	Identity, tax, status, defaults	1	BusinessPartnerId (PK), BPCode (unique), PartyType, LegalName
BusinessPartnerRole	One row per role held	1 : many	BPRoleId, BusinessPartnerId, RoleCode, IsActive, EffectiveFrom, EffectiveTo
BPContact	Named people at the partner	1 : many	BPContactId, BusinessPartnerId, ContactName, Mobile, IsPrimary
BPAddress	Registered, billing, workshop and yard addresses	1 : many	BPAddressId, BusinessPartnerId, AddressType, City, IsPrimary
BPBankAccount	Payment instructions	1 : many	BPBankAccountId, BusinessPartnerId, AccountTitle, IBAN, IsPrimary
BPDocument	Scans: CNIC, licence, NTN, agreements	1 : many	BPDocumentId, BusinessPartnerId, DocTypeId, ExpiryDate, FilePath

Entity	Holds	Cardinality to BP	Key columns
BPDriverDetail	Licence, blood group, employment terms	0..1, exists only with active Driver role	BusinessPartnerId (PK/FK), LicenceNo, LicenceExpiry
BPVendorDetail	Payment terms, supply category	0..1	BusinessPartnerId (PK/FK), PaymentTermDays
BPCustomerDetail	Billing cycle, credit limit, rate basis	0..1	BusinessPartnerId (PK/FK), BillingCycle, CreditLimit

Model rules

- **BR-BP-001** — Role-specific detail rows are created when the role is first activated and are never physically deleted. Deactivating a role sets `BusinessPartnerRole.IsActive = 0` and stamps `EffectiveTo`; the detail row stays for history.
- **BR-BP-002** — Exactly one row per child collection may carry `IsPrimary = 1`. Setting a new primary clears the previous one in the same transaction.
- **BR-BP-003** — `BPCode` is system-generated, unique, and immutable once saved.
- **BR-BP-004** — Bank, Workshop, Body Maker, Tracker Company and Fuel Card Company need no dedicated detail table in V1; their extra attributes fit in the common fields plus contacts and documents. A detail table is added later only if the client confirms role-specific fields for them (see OQ-03).

6. Business Partner — core field specification

These fields apply to every partner regardless of role. They sit on the **General** tab.

#	Field	Type	M/O/C	Validation and behaviour
1	BP Code	nvarchar(20)	Sys	Auto from series BP-{YY}-{000000}. Read-only, shown after save.

#	Field	Type	M/O/C	Validation and behaviour
2	Party Type	enum	M	Person or Company. Radio buttons. Locked once any transaction references the BP.
3	Legal Name	nvarchar(150)	M	Full registered name or CNIC name. Trimmed, min 3 chars. Duplicate-checked (§12).
4	Display / Short Name	nvarchar(60)	O	Used in dropdowns and reports. Defaults to Legal Name truncated to 60.
5	Roles	multi-select	M	At least one role required. Drives conditional tabs (§7).
6	CNIC	nvarchar(15)	C — M when Party Type = Person	Format 000000-00000000-00. Unique across active BPs.
7	NTN	nvarchar(15)	C — M when Party Type = Company	Format per FBR NTN. Unique across active BPs.
8	STRN / Sales Tax No	nvarchar(20)	O	Free text, unique when entered.
9	Filer Status	enum	O	Filer / Non-Filer / Unknown. Default Unknown. Affects withholding reporting later.
10	Primary Mobile	nvarchar(20)	M	Pakistani mobile format 03XX-XXXXXXX or E.164. Used for SMS notifications.
11	Alternate Phone	nvarchar(20)	O	Landline or second mobile.

#	Field	Type	M/O/C	Validation and behaviour
12	Email	nvarchar(100)	C — M when Customer or Bank role active	RFC-valid. Used for document and invoice mail.
13	City	lookup	M	From City master. Type-ahead.
14	Address Line	nvarchar(250)	M	Primary address; also written to BPAddress as the primary Registered row.
15	Branch / Location	lookup	M	Owning branch. Defaults to logged-in user's branch. Controls row-level visibility.
16	Status	enum	Sys on create	Active on save. Changed only through the status action (§13).
17	Opening Balance	decimal(18,2)	O	Default 0.00. Positive = we owe the partner, negative = partner owes us. Posted as an opening entry, not editable after save.
18	Opening Balance Date	date	C — M when Opening Balance ≠ 0	Cannot be a future date.
19	Default Currency	lookup	Sys	PKR in V1, field present for future use, not shown on screen.
20	Notes	nvarchar(1000)	O	Free text, shown on the partner header.
21	Created By / Created On	—	Sys	Logged-in user and server timestamp.

#	Field	Type	M/O/C	Validation and behaviour
22	Modified By / Modified On	—	Sys	Stamped on every save.

Notes on specific fields

- **FR-BP-005** — CNIC and NTN are mutually exclusive by Party Type, but both columns exist. A sole proprietor entered as a Company may carry both; the system does not block that, it only enforces the mandatory one.
- **FR-BP-006** — Opening Balance exists so the client can go live with partners who already have running accounts. If the client prefers to start every partner at zero, this field is hidden by configuration (OQ-02).
- **FR-BP-007** — Branch drives visibility: a user restricted to one branch sees only that branch's partners in lists, but any partner can still be selected on a vehicle if the user has cross-branch permission.

7. Role-specific attribute sets

Each active role adds one collapsible panel to the partner form. An inactive role shows no panel.

7.1 Driver

Field	Type	M/O/C	Rule
Licence Number	nvarchar(30)	M	Unique among active drivers.
Licence Type	lookup	M	LTV / HTV / Motorcycle / Other. HTV required to be assigned to a truck.
Licence Issue Date	date	O	Not future.
Licence Expiry Date	date	M	Must be future at creation. Feeds the expiry notification rule.
Employment Type	enum	M	Employee / Contractor / Ad-hoc.

Field	Type	M/O/C	Rule
Date of Joining	date	C — M when Employee	Not future.
Salary / Monthly Rate	decimal(18,2)	O	Visible only with the Finance permission.
Trip Commission Basis	enum	O	None / Percent of trip income / Fixed per trip.
Commission Value	decimal(18,2)	C — M when basis ≠ None	Percent 0–100 or fixed PKR.
Blood Group	lookup	O	For emergency records.
Emergency Contact Name / Phone	nvarchar(100) / (20)	O	Recommended for HTV drivers.
Guarantor / Reference	nvarchar(150)	O	Free text.
Driver App Access	bit	O	Default off. Switching on requires a linked login (§13).

7.2 Workshop

Field	Type	M/O/C	Rule
Workshop Type	multi-select	M	Engine / Body / Electrical / Tyre / AC / Denting-Painting / General.
Service Rate Basis	enum	O	Per job / Hourly / Contract.
Default Turnaround Days	int	O	Used later for maintenance scheduling.

7.3 Bank

Field	Type	M/O/C	Rule
Branch Name	nvarchar(120)	M	The financing branch.

Field	Type	M/O/C	Rule
Relationship Manager	nvarchar(100)	O	Name; phone goes in Contacts.
Default Finance Type	enum	O	Lease / Ijarah / Loan / Other. Pre-fills the vehicle finance form.
Default Installment Frequency	enum	O	Monthly / Quarterly. Pre-fills the schedule.

7.4 Vendor

Field	Type	M/O/C	Rule
Supply Categories	multi-select	M	Parts / Tyres / Lubricants / Fuel / Toll / Services / Other.
Payment Terms (days)	int	O	0 = cash. Default 0.
Credit Limit	decimal(18,2)	O	Warning only in V1, not a hard block.

7.5 Customer

Field	Type	M/O/C	Rule
Customer Type	enum	M	Adda / Cargo company / Factory / Trader / Other.
Billing Cycle	enum	M	Per trip / Weekly / Fortnightly / Monthly.
Rate Basis	enum	O	Per trip / Per tonne / Per km / Monthly fixed.
Default Rate	decimal(18,2)	O	Pre-fills trip and income entry.
Credit Limit	decimal(18,2)	O	Warning only in V1.
Credit Days	int	O	Drives overdue receivable reporting later.

7.6 Running Customer

The party currently using the vehicle or generating its income. Distinguished from Customer because a vehicle points at exactly one running customer at a time, while Customer is a billing relationship that may cover many vehicles.

Field	Type	M/O/C	Rule
Running Arrangement Type	enum	M	Monthly hire / Per trip / Revenue share.
Agreed Monthly Amount	decimal(18,2)	C — M when Monthly hire	Pre-fills monthly income entry.
Revenue Share %	decimal(5,2)	C — M when Revenue share	0-100.

7.7 Tracker Company, Body Maker, Fuel Card Company

Role	Fields
Tracker Company	Portal URL, Support Contact, Default Monthly Charge per vehicle. All optional in V1; the role exists so Phase 2 tracker integration has a partner to attach to.
Body Maker	Fabrication Types (Container / Tanker / Flatbed / Refrigerated / Other) — mandatory; Default Warranty Months — optional.
Fuel Card Company	Card Program Name — mandatory; Billing Cycle (Weekly / Fortnightly / Monthly) — mandatory; Portal URL — optional. Individual card numbers are captured on the vehicle, not here.

8. Child collections — contacts, addresses, bank accounts, documents

All four are grids on their own tabs with inline add and edit. None is mandatory at creation except the primary address written from the General tab.

8.1 Contacts

Field	Type	M/O/C	Rule
Contact Name	nvarchar(100)	M	—
Designation	nvarchar(60)	O	Owner, Manager, Accounts, Dispatcher.
Mobile	nvarchar(20)	M	Same format as primary mobile.
Email	nvarchar(100)	O	RFC-valid.
Is Primary	bit	O	Only one per partner; setting it clears the previous.
Notes	nvarchar(250)	O	—

8.2 Addresses

Field	Type	M/O/C	Rule
Address Type	enum	M	Registered / Billing / Workshop / Yard / Correspondence.
Address Line 1	nvarchar(250)	M	—
Address Line 2	nvarchar(250)	O	—
City	lookup	M	From City master.
Province	lookup	O	Auto-filled from City where mapped.
Landmark	nvarchar(150)	O	Practical for workshops and addas.
Is Primary	bit	O	One per partner. The General tab address is created as Registered + Primary.

8.3 Bank accounts

Needed for vendor and driver payments and for lease debits.

Field	Type	M/O/C	Rule
Account Title	nvarchar(150)	M	—
Bank Name	nvarchar(100)	M	Free text or Bank BP lookup.
Branch / Code	nvarchar(80)	O	—
Account Number	nvarchar(34)	M	Digits and dashes. Unique per bank within the partner.
IBAN	nvarchar(34)	O	PK + 22 alphanumeric when entered; checksum validated.
Is Primary	bit	O	One per partner.

8.4 Documents

Field	Type	M/O/C	Rule
Document Type	lookup	M	CNIC, Licence, NTN Certificate, Agreement, Cheque copy, Other. From a configurable master.
Document Number	nvarchar(60)	O	—
Issue Date	date	O	Not future.
Expiry Date	date	C — M when the document type is flagged <i>expirable</i>	Feeds expiry notifications.
File	upload	M	PDF, JPG, PNG. Max 10 MB per file. Stored outside the DB with a path reference.
Remarks	nvarchar(250)	O	—

Rules

- **BR-BP-005** — When the Driver role is active, a licence document row is expected. The system warns on save if none is attached but does not block; the warning is configurable to a hard block (OQ-04).
- **BR-BP-006** — Uploaded files are virus-scanned and stored with a generated name; the original filename is kept as metadata.
- **BR-BP-007** — A document row is never deleted once its expiry has driven a notification. It is marked superseded when a renewal is uploaded, and the renewal carries a link to the superseded row.

9. Business Partner screens

9.1 List screen

Default sort: Modified On descending. Page size 25, server-side paging.

Element	Specification
Columns	BP Code, Legal Name, Roles (chips), City, Primary Mobile, Status, Branch, Modified On
Quick search	Single box matching BP Code, Legal Name, Display Name, CNIC, NTN, Mobile — partial match from 3 characters
Filters	Role (multi), Status, City, Branch, Party Type, Has expiring document (next 30 days)
Row actions	View, Edit, Change Status, Duplicate check, View vehicles linked
Header actions	New Business Partner, Export to Excel, Import (Phase 1.1)
Empty state	"No business partners match these filters" with a Clear filters link

9.2 Create / edit screen

A tabbed form, not a wizard. The partner is a reference master that users revisit, and a wizard would slow repeat entry.

Tab	Contents	Visible when
General	§6 core fields, role multi-select	Always
Role Details	One collapsible panel per active role (§7)	At least one role selected
Contacts	Contacts grid	Always
Addresses	Addresses grid	Always
Bank Accounts	Bank accounts grid	Always
Documents	Documents grid	Always
Linked Vehicles	Read-only list of vehicles where this partner is counterparty, bank, body maker or running customer	Edit mode only, and only when at least one link exists
History	Audit entries, role change log, status change log	Edit mode only

9.3 Field behaviour

- **FR-BP-008** — Selecting a role immediately reveals its panel on the Role Details tab without a page reload. Clearing a role hides the panel and warns: values entered there will be retained but inactive.
- **FR-BP-009** — Mandatory-field indicators update live as roles change. A field mandatory only for Driver shows its asterisk only while Driver is selected.
- **FR-BP-010** — Tabs other than General are enabled in create mode and their rows are held client-side until the first save, then persisted together in one transaction. The user never has to save twice to add a contact.
- **FR-BP-011** — Unsaved-change guard: navigating away with dirty state prompts Save, Discard, Cancel.
- **FR-BP-012** — The header of the edit screen always shows BP Code, Legal Name, role chips and Status, whichever tab is open.

10. Business Partner create flow

flowchart TD

```

A[Click New Business Partner] --> B[Pick Person or Company]
B --> C[Enter name, CNIC or NTN, mobile, city]
C --> D{Duplicate match?}
D -- Yes --> E[Show matches, open or continue]
D -- No --> F[Select roles]
E --> F
F --> G[Fill role panels]
G --> H[Add contacts, addresses, documents]
H --> I[Save]
I --> J[BP Code assigned, status Active]

```

Step detail

1. **New** — from the list header, or inline from any partner picker elsewhere in the system (" + Add new" inside the bank dropdown on the vehicle form opens a compact version of this screen with the Bank role pre-selected and locked).
2. **Identity** — Party Type first, because it switches the CNIC/NTN field. Duplicate check fires on blur of Legal Name and of CNIC/NTN (§12).
3. **Roles** — multi-select. At least one required. Roles pre-selected by the calling context cannot be removed in the compact dialog.
4. **Role panels** — only for selected roles; each panel validates independently and shows its own error count in the tab badge.
5. **Child grids** — optional, held client-side until save.
6. **Save** — one server transaction writes BusinessPartner, BusinessPartnerRole rows, detail rows, child rows, the opening balance entry if any, and the audit record. Any failure rolls back everything.

Post-save behaviour

- **FR-BP-013** — On success the screen stays open in edit mode with a toast showing the assigned BP Code, and the Linked Vehicles and History tabs become available.
- **FR-BP-014** — When the partner was created from a picker, the dialog closes and the new partner is selected in the calling field automatically.
- **FR-BP-015** — A **Save and New** action is available for bulk data entry; it retains Party Type, Roles, City and Branch and clears everything else.

- **FR-BP-016** — Documents flagged expirable generate their notification subscriptions at save, not at the next batch run, so a licence expiring in 10 days is picked up immediately.

11. Business Partner business rules

ID	Rule	Enforcement
BR-BP-010	At least one role must be active at all times. Removing the last active role is rejected.	Save-time block
BR-BP-011	A role cannot be deactivated while it is in use: Driver assigned to a vehicle or an open trip; Bank on an open finance agreement; counterparty on an active vehicle relation; Running Customer on an active vehicle.	Save-time block, with a list of the blocking records
BR-BP-012	Party Type cannot change after the first transaction, vehicle link or document is attached.	Field disabled
BR-BP-013	CNIC must be unique across all BPs with status $\neq$ Merged. NTN must be unique across all Company BPs with status $\neq$ Merged.	Unique index + friendly message
BR-BP-014	A BP cannot be set Inactive while any of these is open: driver assignment, vehicle relation, finance agreement, unbilled income or unpaid expense.	Status-change block
BR-BP-015	A BP is never physically deleted. Delete is available only to Admin, only when zero child rows and zero transactions exist, and is recorded as a soft delete with reason.	Permission + guard
BR-BP-016	Adding a role is always allowed and is effective-dated from the date of addition; it does not backdate history.	Automatic

ID	Rule	Enforcement
BR-BP-017	The Driver role requires Licence Number and Licence Expiry. A driver whose licence has expired cannot be assigned to a new trip; the partner record itself stays Active.	Assignment-time block
BR-BP-018	Opening Balance may be entered only once, at creation. Later corrections go through a journal adjustment with reason and audit, not by editing the field.	Field disabled after save
BR-BP-019	Changing Legal Name is allowed but is written to the audit log with old and new values, and the previous name remains searchable.	Audit + search index
BR-BP-020	Blacklisted partners cannot be selected in any picker. Existing links stay intact and existing obligations remain payable.	Picker filter

Role change log

Every add or remove writes a row: BusinessPartnerId, RoleCode, Action (Added / Removed), EffectiveDate, Reason, UserId, Timestamp. This log is shown on the History tab and is the evidence behind BR-BP-011 and BR-BP-016.

12. Validations, duplicate detection and messages

12.1 Duplicate detection

Runs on blur of Legal Name, CNIC, NTN and Primary Mobile, and again on save.

Check	Match type	Response
CNIC exact	Hard	Block save. Offer to open the existing partner.
NTN exact	Hard	Block save. Offer to open the existing partner.

Check	Match type	Response
Mobile exact	Soft	Warn, list the matches, allow continue with a tick box.
Legal Name $\geq$ 85% similar	Soft	Warn with a side-by-side panel of candidates: code, name, roles, city, status.
Name exact + City exact	Soft, higher priority	Warn prominently; this is the common real duplicate.

BR-BP-021 — A soft-duplicate override is recorded in the audit log with the user, the ignored candidate and the timestamp, so duplicates created deliberately can be traced later.

FR-BP-017 — Admin has a *Merge Partners* action: pick survivor and duplicate, the system re-points all children and transactions to the survivor, sets the duplicate to status Merged with a pointer to the survivor, and writes a merge audit record. Merge is irreversible and is deferred to Phase 1.1 if the client prefers (OQ-06).

12.2 Message catalogue

ID	Condition	Message
VAL-BP-001	No role selected	Select at least one role for this business partner.
VAL-BP-002	CNIC format wrong	Enter CNIC as 00000-0000000-0.
VAL-BP-003	CNIC already used	This CNIC belongs to {BPCode} — {LegalName}. Open that record instead.
VAL-BP-004	NTN already used	This NTN belongs to {BPCode} — {LegalName}.
VAL-BP-005	Mobile format wrong	Enter mobile as 03XX-XXXXXXX.
VAL-BP-006	Similar name found	{n} partners have a similar name in {City}. Review before saving.
VAL-BP-007	Licence expiry in the past	Licence expiry must be a future date.

ID	Condition	Message
VAL-BP-008	Driver role without licence number	Licence number is required for the Driver role.
VAL-BP-009	Last role being removed	A business partner must keep at least one role.
VAL-BP-010	Role in use on removal	{Role} cannot be removed — it is in use on {n} records. View list.
VAL-BP-011	Opening balance without date	Enter the opening balance date.
VAL-BP-012	Opening balance date in future	Opening balance date cannot be in the future.
VAL-BP-013	Email missing for Customer role	Email is required for customers.
VAL-BP-014	Document expirable without expiry	Enter the expiry date for {DocumentType}.
VAL-BP-015	File too large	File exceeds 10 MB. Compress it or upload a smaller scan.
VAL-BP-016	Deactivation blocked	This partner has {n} open records and cannot be deactivated. View list.

All messages are held in a resource file so they can be translated or reworded without a code change.

13. Status, audit and permissions

13.1 Status lifecycle

```
stateDiagram-v2
    [*] --> Active: Save
    Active --> Inactive: Deactivate
    Inactive --> Active: Reactivate
    Active --> Blacklisted: Blacklist
    Blacklisted --> Active: Remove blacklist
    Active --> Merged: Merge into survivor
    Inactive --> Merged: Merge into survivor
```

Status	Selectable in pickers	Editable	Notes
Active	Yes	Yes	Default on save
Inactive	No	Yes	Existing links stay; blocked by BR-BP-014 while anything is open
Blacklisted	No	Yes	Requires a reason, minimum 10 characters
Merged	No	No	Read-only tombstone pointing at the survivor

Every status change captures status from, status to, reason, effective date, user and timestamp.

13.2 Audit

Written for: create, every field change on the General and role panels, role add/remove, status change, child row add/edit/delete, document upload and supersede, duplicate override, merge.

Each entry holds: entity, record id, action, field, old value, new value, reason where captured, user id, user name, server timestamp, IP or device.

BR-BP-022 — Audit rows are append-only. No role, including Admin, can edit or delete them through the application.

13.3 Permissions

Capability	Admin	Fleet Manager	Finance	Operations	Read Only
View partners	Yes	Yes	Yes	Yes	Yes
Create / edit partner	Yes	Yes	Yes	Limited to Customer and Running Customer roles	No
View salary, commission, credit limit, opening balance	Yes	No	Yes	No	No
Add or remove roles	Yes	Yes	No	No	No
Change status / blacklist	Yes	No	No	No	No
Merge partners	Yes	No	No	No	No
Upload documents	Yes	Yes	Yes	Yes	No
Grant Driver App access	Yes	Yes	No	No	No
Export list	Yes	Yes	Yes	No	Yes

Roles and their capabilities are configurable per the source document; this matrix is the shipped default.

Module B — Vehicle

14. Vehicle creation objectives

Vehicle creation is not just a master record. It is the opening entry of the vehicle's financial and commercial life, so the screen captures four things at once:

1. **Identity and technical data** — registration number, make, model, engine, chassis, capacity.
2. **Commercial relationship** — which of the five categories applies, and which Business Partner is on the other side of it.
3. **Opening financial position** — purchase price, amount paid at creation, and the bank lease agreement with its installment schedule where applicable.
4. **Attached items and documents** — container, AC, tyres, registration book, insurance, fitness, permits.

The requirement in §8 of the source document — that a vehicle created with PKR 2,000,000 paid and later PKR 150,000 of installments shows PKR 2,150,000 paid — is only achievable if the initial payment is posted as a transaction at creation rather than stored as a number on the vehicle. That is the central design decision of this module.

Lifecycle entry

```

flowchart LR
    N[New vehicle form] --> D[Draft]
    D --> A[Active]
    A --> AS[Assigned / Operational]
    AS --> M[Maintenance]
    M --> AS
    A --> R[Retired / Sold / Transferred]

```

FR-VH-001 — A vehicle may be saved as **Draft** with only its identity fields, so a purchase in progress can be recorded. It becomes **Active** only when category, counterparty and acquisition data are complete. Draft vehicles cannot be assigned, cannot take expenses and do not appear in operational pickers.

15. Vehicle data model

One save of the vehicle creation screen can write to eight tables.

```

flowchart TD
    V[Vehicle] --> REL[VehicleBusinessPartnerRelation]
    V --> LIFE[VehicleLifecycleHistory]
    V --> TXN[VehicleTransaction]
    V --> ITEM[VehicleAttachedItem]
    V --> VDOC[VehicleDocument]
    V --> ODO[OdometerReading]
    V --> FIN[VehicleFinanceAgreement]
    FIN --> INST[VehicleInstallment]
    REL --> BP[BusinessPartner]
    FIN --> BP

```

Entity	Written at creation	Rows	Purpose
Vehicle	Always	1	Identity, technical data, current status, current category (denormalised for speed)
VehicleBusinessPartnerRelation	When category $\neq$ Self Owned	1	Dated link to the counterparty. Closed and superseded on category change, never edited
VehicleLifecycleHistory	Always	1-2	Created, and Activated if not saved as Draft
VehicleTransaction	When purchase price or initial payment entered	1-2	Acquisition and initial payment postings (§19)
VehicleFinanceAgreement	When Finance Type = Bank Lease	1	Bank, agreement reference, amounts, schedule terms
VehicleInstallment	When Finance Type = Bank Lease	n = tenure	The full expected schedule, status Pending
VehicleAttachedItem	When items entered	0..n	Container, AC, tyre set, crane
VehicleDocument	When documents uploaded	0..n	Registration, insurance, fitness, permit, lease
OdometerReading	When opening odometer entered	0..1	Source = Vehicle Creation
VehicleRecurringCharge	When periodic charges are configured	0..n	Insurance, tracker fee, rent and other recurring obligations (§19A)

Entity	Written at creation	Rows	Purpose
VehicleRecurringChargeEntry	No	—	Due and paid entries produced later by the nightly generation job

Model rules

- **BR-VH-001** — `Vehicle.CurrentCategory` and `Vehicle.CurrentCounterpartyId` are convenience copies of the open `VehicleBusinessPartnerRelation`. The relation table is the record of truth; the copies are maintained by the same transaction and are never edited directly.
- **BR-VH-002** — A vehicle has at most one open relation at a time, identified by `EffectiveTo IS NULL`. Enforced by a filtered unique index.
- **BR-VH-003** — All financial figures shown on the vehicle screen are aggregates over `VehicleTransaction`. No running-total column is stored on Vehicle.
- **BR-VH-004** — A vehicle has at most one active finance agreement. Historic agreements remain with status Closed or Settled.

16. Vehicle — core field specification

16.1 Identity

#	Field	Type	M/O/C	Validation and behaviour
1	Vehicle Code	nvarchar(20)	Sys	Series <code>VH-{YY}-{0000}</code> . Internal key shown on all screens.
2	Registration Number	nvarchar(20)	M	Unique among non-retired vehicles. Stored uppercase, spaces and dashes stripped for comparison so <code>LES-1234</code> and <code>LES 1234</code> clash.
3	Registration City	lookup	O	Province/city of registration.
4	Chassis Number	nvarchar(30)	O	Unique when entered.
5	Engine Number	nvarchar(30)	O	Unique when entered.

#	Field	Type	M/O/C	Validation and behaviour
6	Vehicle Type	lookup	M	Truck / Trailer / Prime Mover / Pickup / Tanker / Bus / Car / Bike / Other. Configurable master.
7	Make	lookup	M	Hino, Isuzu, Nissan, Mercedes, Master, Toyota, Suzuki, Other. Configurable.
8	Model	nvarchar(60)	M	Free text or filtered lookup by Make.
9	Manufacturing Year	int	O	1950 to current year + 1.
10	Colour	nvarchar(30)	O	—

16.2 Technical and capacity

#	Field	Type	M/O/C	Validation and behaviour
11	Fuel Type	enum	M	Diesel / Petrol / CNG / LPG / Hybrid / Electric. Drives fuel entry units.
12	Tank Capacity (litres)	decimal(10,2)	O	Used to sanity-check fuel entries.
13	Load Capacity	decimal(10,2)	C — M for Truck, Trailer, Tanker, Prime Mover	Value plus unit.
14	Capacity Unit	enum	C — M when Load Capacity entered	Tonne / Kg / Litre / CFT / Passengers.
15	Axle Configuration	lookup	O	4×2, 6×4, 10-wheeler, 22-wheeler.
16	Body Type	lookup	O	Open / Container / Tanker / Flatbed / Refrigerated / Dumper.
17	Tyre Count	int	O	2-22. Used by the tyre-set attached item.

#	Field	Type	M/O/C	Validation and behaviour
18	GVW	decimal(10,2)	O	Gross vehicle weight from the registration book.

16.3 Operational

#	Field	Type	M/O/C	Validation and behaviour
19	Branch / Location	lookup	M	Defaults to the user's branch. Drives visibility.
20	Opening Odometer (km)	int	O	Reading on the date the vehicle enters the system. Written to OdometerReading. Later readings cannot be lower.
21	Opening Odometer Date	date	C — M when opening odometer entered	Not future.
22	Default Driver	BP picker, Driver role	O	Optional at creation; a formal assignment record is created if set.
23	Fuel Card Company	BP picker, Fuel Card role	O	—
24	Fuel Card Number	nvarchar(30)	C — M when fuel card company set	Unique among active vehicles.
25	Tracker Company	BP picker, Tracker role	O	Phase 2 uses this; captured now.
26	Tracker Device ID	nvarchar(40)	O	Free text in V1.
27	Status	enum	Sys	Draft or Active on save (FR-VH-001).

#	Field	Type	M/O/C	Validation and behaviour
28	Remarks	nvarchar(1000)	O	—
29	Created By / On, Modified By / On	—	Sys	Stamped automatically.

FR-VH-002 — Registration Number is the field users search by, so it is the first field on the form and the primary column everywhere. Chassis and engine numbers are for verification, not day-to-day use.

FR-VH-003 — Vehicle Type drives conditional mandatory fields (load capacity) and the attached-item suggestions offered in §20.

17. Ownership category specification

Selecting the category changes which fields appear below it. Every category except Self Owned requires a counterparty and creates a `VehicleBusinessPartnerRelation`.

Category	Counterparty role required	Extra fields	Creates relation
Self Owned	None	Purchase date, purchase price, seller (Vendor, optional)	No
Shared	Any BP	Partner, share %, arrangement basis, agreement ref, start date, expense-sharing rule	Yes
Rented	Any BP	Lessor, monthly rent, security deposit, rent due day, start date, end date, agreement ref	Yes
Customer Arrangement	Customer or Running Customer	Customer, arrangement type, agreed amount, start date, end date, agreement ref	Yes
Bank Leased	Bank	Full finance block (§18)	Yes

17.1 Shared

Field	Type	M/O/C	Rule
Sharing Partner	BP picker	M	Any active BP.
Our Share %	decimal(5,2)	M	0.01-100. Partner share displayed as 100 – our share.
Sharing Basis	enum	M	Profit share / Revenue share / Fixed monthly.
Fixed Monthly Amount	decimal(18,2)	C — M when basis = Fixed monthly	—
Expense Sharing Rule	enum	M	All expenses shared in the same ratio / Only major expenses shared / Each party bears its own. Drives P&L allocation later.
Agreement Reference	nvarchar(60)	O	—
Start Date	date	M	Not future beyond today.
End Date	date	O	Must be after start date.

17.2 Rented

Field	Type	M/O/C	Rule
Lessor	BP picker	M	—
Rent Amount	decimal(18,2)	M	Greater than zero.
Rent Frequency	enum	M	Monthly / Weekly / Per trip.
Rent Due Day	int	C — M when frequency = Monthly	1-31; 29-31 fall back to month end.
Security Deposit	decimal(18,2)	O	Posted as a transaction if entered (§19).

Field	Type	M/O/C	Rule
Start Date / End Date	date	M / O	End after start.
Agreement Reference	nvarchar(60)	O	—

17.3 Customer Arrangement

Field	Type	M/O/C	Rule
Customer	BP picker, Customer or Running Customer role	M	—
Arrangement Type	enum	M	Dedicated monthly / Per trip / Revenue share.
Agreed Amount	decimal(18,2)	C — M for Dedicated monthly	Pre-fills monthly income entry.
Revenue Share %	decimal(5,2)	C — M for Revenue share	0-100.
Start Date / End Date	date	M / O	End after start.
Agreement Reference	nvarchar(60)	O	—

Category change rules

- **BR-VH-005** — Changing category is a separate action on the vehicle screen, not an edit of the creation form. It closes the open relation with `EffectiveTo = change date - 1`, opens a new relation from the change date, writes a lifecycle history row with reason, and leaves all prior transactions untouched.
- **BR-VH-006** — The new relation's start date cannot overlap the closed one, and cannot precede the vehicle's acquisition date.

- **BR-VH-007** — Category cannot be changed to anything other than Bank Leased while an active finance agreement with outstanding balance exists. Settle or close the agreement first.
- **BR-VH-008** — Share percentages and rent amounts are versioned with the relation. Changing them mid-arrangement closes the relation and opens a new one, so historic P&L stays correct.

18. Acquisition and finance capture

18.1 Acquisition block — every category

Field	Type	M/O/C	Rule
Acquisition Date	date	M	Date the vehicle entered the fleet. Not future. No transaction may predate it.
Acquisition Type	enum	M	Purchase / Lease / Rent / Shared induction / Customer induction. Defaulted from the category, editable.
Seller / Source	BP picker	O	Vendor or any BP.
Purchase Price	decimal(18,2)	C — M for Self Owned and Bank Leased	Greater than zero.
Amount Paid at Creation	decimal(18,2)	C — M when purchase price entered	0 to purchase price. This is the figure in the §8 worked example.
Payment Mode	enum	C — M when amount paid > 0	Cash / Bank transfer / Cheque / Pay order.
Payment Reference	nvarchar(60)	O	Cheque or transaction number.
Registration Cost	decimal(18,2)	O	Posted separately as a major expense (§19).

18.2 Finance block — Bank Leased only

Field	Type	M/O/C	Rule
Finance Type	enum	M	Fully Paid / Bank Lease / Ijarah / Loan. Configurable list per source §8.
Bank	BP picker, Bank role	M	Only active Bank-role partners.
Agreement Number	nvarchar(60)	M	Unique per bank.
Agreement Date	date	M	Not after acquisition date + 90 days.
Finance Amount	decimal(18,2)	M	Purchase price – down payment, editable; warn if it does not reconcile.
Down Payment	decimal(18,2)	M	Ties to Amount Paid at Creation; the two are cross-checked.
Installment Amount	decimal(18,2)	M	Greater than zero.
Frequency	enum	M	Monthly / Quarterly / Half-yearly.
Tenure (installments)	int	M	1–120.
First Installment Due Date	date	M	On or after agreement date.
Markup / Profit Rate %	decimal(5,2)	O	Recorded for reference; V1 does not split principal and markup (OQ-07).
Total Payable	decimal(18,2)	Sys	Installment × tenure, displayed live as the user types.
Residual / Balloon Amount	decimal(18,2)	O	Due at the end of the tenure; added to the schedule as a final row.
Security Deposit	decimal(18,2)	O	Posted as a transaction.

18.3 Schedule generation

FR-VH-004 — On save the system generates `VehicleInstallment` rows: one per tenure period, each with installment number, due date, expected amount, status Pending, and zero paid amount. The due date advances by the frequency from the first due date; a due day of 29–31 falls back to the last day of shorter months.

FR-VH-005 — The generated schedule is shown to the user before save in a preview grid, with an editable due-date column, so irregular bank schedules can be corrected without a developer.

FR-VH-006 — Outstanding is always computed as *total payable + residual – sum of paid installment amounts*. It is never stored.

BR-VH-009 — Down payment and Amount Paid at Creation must be equal when Finance Type = Bank Lease. If the user enters different figures the save is blocked with VAL-VH-012; this prevents the paid-to-date figure being counted twice.

BR-VH-010 — Finance Amount + Down Payment should equal Purchase Price. A mismatch produces a warning, not a block, because banks often finance a valuation that differs from the invoice.

19. Opening postings — what the system writes by itself

The user types amounts once, on the creation form. The system converts them into vehicle transactions so that every later total is a sum, never a stored figure.

Trigger on the form	Transaction type	Amount	Business Partner	Reference
Purchase Price entered	Acquisition	Purchase price	Seller	Vehicle acquisition – {RegNo}
Amount Paid at Creation > 0	Initial Payment	Amount paid	Seller or Bank	Initial payment – {PaymentMode} {Reference}
Registration Cost > 0	Major Expense, subtype Registration	Registration cost	Seller or authority	Registration – {RegNo}
Security Deposit > 0 (Rented or Bank Leased)	Deposit	Deposit amount	Lessor or Bank	Security deposit
Attached item with cost > 0	Major Expense, subtype from item type	Item cost	Supplier or Body Maker	{ItemType} – {Description}

Trigger on the form	Transaction type	Amount	Business Partner	Reference
Bank Lease saved	No transaction	—	—	The schedule is an expectation, not a posting. Installments post when paid.

All opening postings carry: transaction date = acquisition date (or the item's installation date where later), created-by = logged-in user, created-on = server timestamp, source = `VehicleCreation`, and a link back to the vehicle.

The §8 worked example

A vehicle is created with purchase price PKR 5,000,000, amount paid at creation PKR 2,000,000, and a bank lease for the balance. Later, one installment of PKR 150,000 is recorded.

Event	Posting	Paid-to-date shown on the vehicle screen
Creation	Acquisition 5,000,000; Initial Payment 2,000,000	PKR 2,000,000
Installment 1 recorded	Installment 150,000	PKR 2,150,000
Installment 2 recorded	Installment 150,000	PKR 2,300,000

Paid-to-date = sum of Initial Payment + Installment transactions. Outstanding financing = total payable + residual – installments paid. Both are derived, so a correction to any single transaction flows through instantly.

BR-VH-011 — Opening postings are marked `IsSystemGenerated = 1`. They cannot be deleted. Correcting them requires the controlled adjustment route in source §13, which posts a reversing entry with reason and audit.

BR-VH-012 — If the vehicle is saved as Draft, no postings are written. They are created when the vehicle is activated, dated by the acquisition date, not the activation date.

19A. Recurring vehicle charges

The answer is both, in sequence. The system generates the entry automatically on its due date, but it arrives as **Due** — an expectation, not a payment. A user confirms it with the actual amount and date, and only that confirmation posts a transaction to the vehicle ledger. Auto-posting straight to the ledger is available per charge, but is off by default.

The reason is that an unconfirmed auto-posted expense corrupts the vehicle's profit figure. An insurance premium that was never actually paid, or was paid at a renegotiated amount, would sit in the P&L as fact. A Due entry that nobody clears is visible as an outstanding item, which is the correct picture.

19A.1 Configuration — VehicleRecurringCharge

One row per recurring obligation on a vehicle. Set up at vehicle creation (a new step-4 grid) or added later from the vehicle screen.

Field	Type	M/O/C	Rule
Charge Type	lookup	M	Bank Installment, Insurance Premium, Tracker Fee, Token Tax, Route Permit, Fitness Renewal, Fuel Card Fee, Vehicle Rent Payable, Maintenance Contract, Shared Partner Payout, Driver Salary, Other. Configurable master.
Payee	BP picker	M	Filtered by the role implied by the charge type — Bank for installments, Tracker Company for tracker fees.
Expense Type mapping	lookup	M	Which expense type the posted transaction carries. Defaulted from charge type.
Amount	decimal(18,2)	C — M unless Amount Basis = Variable	The expected periodic amount.

Field	Type	M/O/C	Rule
Amount Basis	enum	M	Fixed / Variable / Percentage of income. Variable generates a Due entry with the last paid amount pre-filled.
Frequency	enum	M	Monthly / Quarterly / Half-yearly / Yearly / Weekly / Custom days.
Due Day or Date	int / date	M	Day of month for monthly; anniversary date for yearly. 29–31 falls back to month end.
Start Date	date	M	Not before the vehicle's acquisition date.
End Date	date	O	Blank = open-ended until the vehicle is disposed.
Number of Occurrences	int	O	Alternative to end date. Mutually exclusive with it.
Posting Mode	enum	M	Generate as Due (default) / Auto-post / Reminder only .
Generate Lead Days	int	O	How many days before the due date the entry is created. Default 7, configurable.
Reminder Rule	lookup	O	Which notification rule fires. Defaults to the charge type's rule.
Tax / Withholding %	decimal(5,2)	O	Recorded on the generated entry; not calculated into the ledger in V1.
Is Active	bit	Sys	Active on save. Suspended automatically on vehicle disposal.
Last Generated Period	—	Sys	Idempotency key for the generation job.

Field	Type	M/O/C	Rule
Next Due Date	—	Sys	Derived, shown on the vehicle screen.

19A.2 Posting modes

Mode	Generates	Posts to ledger	Use it for
Generate as Due	A Due entry on the due date, minus lead days	Only on user confirmation	Insurance, tracker fees, token tax, rent — anything whose amount or timing can move
Auto-post	A Due entry and immediately posts the transaction	Yes, without confirmation	A genuinely fixed, always-paid charge the client trusts — a fixed tracker fee debited automatically by the vendor
Reminder only	A notification, no entry at all	No	Renewals the client wants to be warned about but records elsewhere

BR-VH-027 — Auto-post is permitted only when Amount Basis = Fixed. A Variable charge cannot be set to Auto-post; the form blocks it.

BR-VH-028 — Enabling Auto-post is an Admin-only permission, and every auto-posted transaction is stamped `IsSystemGenerated = 1` and `Source = RecurringCharge`, so an audit can separate it from user-entered expenses.

19A.3 Bank installments are not duplicated

BR-VH-029 — When a vehicle has an active finance agreement, its installment schedule (§18.3) is the recurring charge for Bank Installment. The generation engine reads `VehicleInstallment` rows and surfaces them as Due; it does not create a second parallel schedule. A user may not add a manual Bank Installment recurring charge to a vehicle that already has an active agreement — VAL-VH-019.

The practical effect: one schedule, two views. Finance sees it as a lease schedule; the payables workbench sees it as this month's due items alongside insurance and tracker fees.

19A.4 Generated entry lifecycle

stateDiagram-v2

```

[*] --> Scheduled: Charge configured
Scheduled --> Due: Generation job, due date minus lead days
Due --> Overdue: Due date passed, unconfirmed
Due --> Paid: User confirms, transaction posted
Overdue --> Paid: User confirms, transaction posted
Due --> Waived: Admin waives with reason
Overdue --> Waived: Admin waives with reason
Scheduled --> Cancelled: Charge ended or vehicle disposed

```

The generated entry holds: vehicle, charge, period key, due date, expected amount, status, actual paid date, actual amount, payment mode, reference, payee, remarks, linked transaction id, generated-by (system), confirmed-by user.

ID	Rule
BR-VH-030	The generation job runs nightly and is idempotent on (ChargeId, PeriodKey). Re-running it never creates a duplicate entry, so a job restart is safe.
BR-VH-031	Confirming an entry posts exactly one <code>VehicleTransaction</code> with the mapped expense type and the actual amount, linked back to the entry. Paid amount and expected amount are both retained, so variance is reportable.
BR-VH-032	A Paid entry cannot be deleted. It is reversed through the controlled adjustment route (source §13), which also resets the entry to Due.
BR-VH-033	A charge whose amount changes is not edited in place. The current row is end-dated and a new row starts from the effective date, so historic entries keep the amount that actually applied.

ID	Rule
BR-VH-034	When a vehicle moves to Sold, Retired or Transferred, every active charge is end-dated at the disposal date and unconfirmed future entries are cancelled. Overdue unpaid entries remain, because the money is still owed.
BR-VH-035	When a vehicle's category changes, charges tied to the closed relation — rent payable on a Rented vehicle, partner payout on a Shared one — are end-dated at the relation's close date. The user is prompted to set up replacements.
BR-VH-036	A user may add a one-off entry manually against any charge, for an out-of-cycle payment, without disturbing the schedule.

19A.5 Screens

Screen	Contents
Vehicle screen → Recurring Charges tab	Grid of configured charges: type, payee, amount, frequency, next due, posting mode, status. Add, end-date, view generated history.
Vehicle screen → Due & Payments panel	This vehicle's Due and Overdue entries with a Confirm Payment action
Finance → Payables Due workbench	Fleet-wide list of Due and Overdue entries across all vehicles and charge types. Filters by charge type, payee, branch, due window. Bulk confirm with a common payment date and mode — this is how a user clears twelve tracker fees in one action
Dashboard tile	Count and total of items due in the next 7 days, and overdue count

FR-VH-016 — The payables workbench is the primary daily screen for the Finance user. Entering payments vehicle by vehicle is possible but not the intended path.

19A.6 Notification hooks

Event	Default lead	Rule
Entry generated and becoming due	3 days before due date	Configurable per charge type, per source §9
Entry overdue	1 day after due date, then every 3 days	Configurable, with an escalation recipient
Charge schedule ending	30 days before end date	Prompts renewal setup

OQ-12 — Which charge types should ship with Auto-post enabled by default, if any? The recommendation is none.

OQ-13 — Is insurance paid annually in one premium, or in installments? If installments, it is configured as a monthly or quarterly recurring charge rather than a yearly one.

OQ-14 — Should driver salary be a vehicle-level recurring charge, or does it belong to the driver and get allocated to vehicles later? It is listed as a charge type above, but that is a placeholder pending this answer.

20. Attached items and documents at creation

20.1 Attached items

Major removable assets fitted to the vehicle. They are separated from expenses because they have a life of their own: a container can be moved to another vehicle, a tyre set replaced, an AC unit removed on sale.

Field	Type	M/O/C	Rule
Item Type	lookup	M	Container, AC Unit, Tyre Set, Crane, Tanker Body, Refrigeration Unit, Tail Lift, Tracker Device, Other. Configurable.
Description	nvarchar(150)	M	Size, spec, brand.
Serial / Identification Number	nvarchar(60)	O	Unique when entered. Container number for containers.

Field	Type	M/O/C	Rule
Supplier / Body Maker	BP picker	O	Vendor or Body Maker role.
Installation Date	date	M	Not before the vehicle's acquisition date, not future.
Cost	decimal(18,2)	O	Posts a major expense when greater than zero (§19).
Expected Life / Warranty Until	date	O	Feeds replacement reminders.
Condition	enum	O	New / Used / Refurbished.
Status	enum	Sys	Attached on creation.

- **FR-VH-007** — Item Type suggestions are filtered by Vehicle Type: a Tanker offers Tanker Body and Refrigeration Unit first; a Prime Mover offers Container and Crane.
- **BR-VH-013** — An attached item can later be detached with a date and reason, or transferred to another vehicle. A transfer closes the item on the source vehicle and opens it on the target with the same serial number, preserving the chain.
- **BR-VH-014** — Serial numbers are unique among attached items with status Attached. The same container number may reappear after the earlier one is detached.

20.2 Documents

Field	Type	M/O/C	Rule
Document Type	lookup	M	Registration Book, Insurance Policy, Fitness Certificate, Route Permit, Token Tax, Lease Agreement, Purchase Invoice, Other. Configurable, each flagged expirable or not.
Document Number	nvarchar(60)	C — M for Insurance, Fitness, Permit	—

Field	Type	M/O/C	Rule
Issuing Authority / Provider	nvarchar(120) or BP picker	O	Insurance company as a Vendor BP where available.
Issue Date	date	O	Not future.
Expiry Date	date	C — M when the type is expirable	Must be future at upload.
Amount / Premium	decimal(18,2)	O	Posts a major expense if greater than zero and the user ticks <i>post as expense</i> .
File	upload	M	PDF, JPG, PNG, max 10 MB.
Remarks	nvarchar(250)	O	—

- **FR-VH-008** — Saving a document with an expiry date creates its notification subscription immediately, using the rule configured by Admin (source §9). A fitness certificate expiring in 20 days with a 30-day rule alerts on the next notification run.
- **BR-VH-015** — A vehicle cannot be activated without a Registration Book document when the category is Self Owned or Bank Leased. For Rented and Customer Arrangement it is a warning, since the papers sit with the owner.
- **BR-VH-016** — Renewals are uploaded as new rows referencing the superseded document. The old row stays with status Superseded and its own expiry history.

21. Vehicle creation screen

Unlike the partner form, vehicle creation is a **five-step wizard**. It is a low-frequency, high-value entry with conditional branches and financial consequences, and a wizard keeps the finance step from being skipped.

Step	Title	Contents	Can save as Draft
1	Vehicle details	§16.1 identity + §16.2 technical	Yes, from here on

Step	Title	Contents	Can save as Draft
2	Ownership	Category, counterparty, category-specific fields (§17)	Yes
3	Acquisition & finance	§18.1 acquisition, §18.2 finance block, schedule preview	Yes
4	Items & documents	§20 attached items grid, documents grid, §19A recurring charges grid	Yes
5	Review & activate	Read-only summary of everything, list of postings about to be written, Activate button	—

Screen rules

- **FR-VH-009** — Steps 2 to 5 can be reached only after step 1 validates. The wizard shows a progress bar with per-step error counts.
- **FR-VH-010** — Step 3's finance block appears only when Category = Bank Leased or Finance Type ≠ Fully Paid. For Rented and Customer Arrangement the block is replaced by the rent or arrangement summary from step 2.
- **FR-VH-011** — Step 5 lists every transaction the save will create, with type and amount, before the user commits. This is the user's last chance to catch a mistyped purchase price.
- **FR-VH-012** — **Save Draft** is available from step 1 onward and writes only the vehicle row with status Draft. **Activate** is available only from step 5 and writes everything in one transaction.
- **FR-VH-013** — A partner needed on any step can be created inline through the compact partner dialog (§10) without leaving the wizard or losing entered data.
- **FR-VH-014** — **Copy from existing vehicle** on step 1 pre-fills type, make, model, capacity, axle and body from a chosen vehicle, leaving registration, chassis and engine blank. Useful when a fleet buys five identical trucks.

Vehicle list screen

Element	Specification
Columns	Reg No, Type, Make/Model, Category, Counterparty, Branch, Status, Current Driver, Next document expiry
Filters	Category, Status, Type, Branch, Make, Finance type, Document expiring in 30 days, Has open finance
Quick search	Reg No, chassis, engine, vehicle code
Row actions	Open vehicle screen, Change category, Assign driver, Add expense, Add income
Header actions	New Vehicle, Export, Bulk document upload (Phase 1.1)

22. Vehicle business rules and messages

22.1 Business rules

ID	Rule	Enforcement
BR-VH-017	Registration Number is unique among vehicles not in status Sold or Transferred. A re-registered vehicle updates its own number; the old number is kept in history.	Filtered unique index
BR-VH-018	Acquisition date cannot be in the future, and no transaction, document issue date or item installation date may precede it.	Save-time block

ID	Rule	Enforcement
BR-VH-019	A vehicle cannot be activated unless: category chosen, counterparty present where the category requires it, acquisition date and type set, and the §20 document rule satisfied.	Activate blocked, with a checklist of what is missing
BR-VH-020	The counterparty picker is filtered by role and excludes Inactive, Blacklisted and Merged partners.	Picker filter
BR-VH-021	Amount Paid at Creation cannot exceed Purchase Price.	Field validation
BR-VH-022	A driver can be the default driver of only one Active vehicle at a time. Assigning him elsewhere prompts to release the current assignment.	Save-time prompt
BR-VH-023	A vehicle is never deleted once it has any transaction. It is retired, sold or transferred, each with date, reason and counterparty.	Guard
BR-VH-024	A Draft vehicle with no transactions may be deleted by Admin within 30 days, recorded as a soft delete.	Permission + guard
BR-VH-025	Opening odometer is the floor for all later readings on that vehicle.	Validation at every odometer entry
BR-VH-026	Fuel card number must be unique among Active vehicles; reassigning it prompts to release it from the current vehicle.	Save-time prompt

22.2 Message catalogue

ID	Condition	Message
VAL-VH-001	Registration number blank	Enter the registration number.

ID	Condition	Message
VAL-VH-002	Registration number exists	{RegNo} is already registered as {VehicleCode}. Open that vehicle.
VAL-VH-003	Chassis or engine duplicate	This {field} is already recorded on {VehicleCode}.
VAL-VH-004	Load capacity missing for a truck	Load capacity is required for {VehicleType}.
VAL-VH-005	Acquisition date in future	Acquisition date cannot be in the future.
VAL-VH-006	Counterparty missing	Select the {Bank / Lessor / Customer / Sharing partner} for this category.
VAL-VH-007	Counterparty lacks the role	{PartnerName} does not have the {Role} role. Add the role or pick another partner.
VAL-VH-008	Amount paid exceeds price	Amount paid cannot be more than the purchase price.
VAL-VH-009	Share % out of range	Share percentage must be between 0.01 and 100.
VAL-VH-010	End date before start	End date must be after the start date.
VAL-VH-011	Installment amount or tenure zero	Enter the installment amount and tenure to generate the schedule.
VAL-VH-012	Down payment ≠ amount paid	Down payment ({a}) must match the amount paid at creation ({b}).
VAL-VH-013	Finance amount does not reconcile — warning	Finance amount plus down payment is {x}, purchase price is {y}. Continue anyway?
VAL-VH-014	Activation without registration book	Upload the registration book before activating this vehicle.
VAL-VH-015	Item installed before acquisition	Installation date cannot be before the acquisition date.

ID	Condition	Message
VAL-VH-016	Document expiry in the past	{DocumentType} has already expired. Upload a current document.
VAL-VH-017	Driver already assigned	{DriverName} is the default driver of {RegNo}. Release that assignment?
VAL-VH-018	Fuel card in use	This fuel card is assigned to {RegNo}. Reassign it?

23. Vehicle status, audit, permissions and notification hooks

23.1 Status values

Status	Meaning	Can take transactions	Appears in operational pickers
Draft	Being entered, incomplete	No	No
Active	In the fleet, available	Yes	Yes
Assigned	Active with a driver and open trip	Yes	Yes
Under Maintenance	In workshop	Yes, expenses only	No
Temporarily Unavailable	Off road, documents expired, impounded	Yes, expenses only	No
Retired	Withdrawn, still owned	No new, historic visible	No
Sold	Disposed, with buyer and sale amount	No new	No
Transferred	Moved to another party or branch	No new	No

Every status change writes a `VehicleLifecycleHistory` row: from status, to status, effective date, reason, reference, user, timestamp — as required by source §4.

23.2 Audit at creation

One creation writes audit entries for: vehicle created, relation opened, finance agreement created, schedule generated (count of rows), each opening posting, each item attached, each document uploaded, and status set to Active. The History tab replays this as a single dated group so a reviewer sees the whole induction at once.

23.3 Permissions

Capability	Admin	Fleet Manager	Finance	Operations	Driver	Read Only
View vehicle list and screen	Yes	Yes	Yes	Yes	Own vehicle only	Yes
Create vehicle (Draft)	Yes	Yes	No	No	No	No
Enter acquisition and finance data	Yes	No	Yes	No	No	No
Activate vehicle	Yes	Yes	No	No	No	No
View purchase price and finance figures	Yes	Yes	Yes	No	No	No
Change category	Yes	No	No	No	No	No
Attach or detach items	Yes	Yes	No	No	No	No
Upload vehicle documents	Yes	Yes	Yes	Yes	No	No
Retire / sell / transfer	Yes	No	No	No	No	No

FR-VH-015 — Fleet Manager can create a Draft vehicle with technical data but cannot enter or see the acquisition and finance step; Finance completes it. This split is configurable — if the client wants one user to do the whole induction, grant both capabilities to one role. §23B defines how.

23.4 Notification hooks created at vehicle creation

Source	Rule consumed	Default lead time
First installment due date	Bank installment due	3 days, configurable per source §9
Insurance expiry	Document expiry	30 days
Fitness certificate expiry	Document expiry	30 days

Source	Rule consumed	Default lead time
Route permit expiry	Document expiry	30 days
Token tax expiry	Document expiry	30 days
Lease agreement end date	Contract expiry	60 days
Rent due day (Rented)	Rent due	3 days
Attached item warranty end	Maintenance due	15 days

All lead times are values in the notification rule master, not constants in code.

Cross-cutting

23A. Document management

One document engine serves both masters. The Documents tab on a partner (§8.4) and on a vehicle (§20.2) are two entry points into the same storage, versioning, expiry and notification machinery, so a fitness certificate and a driving licence behave identically.

Most documents in this business are periodic: insurance renews yearly, fitness every six or twelve months, route permits annually, token tax annually. The engine therefore treats a document not as a file but as a **slot that holds a current version and a history of superseded ones**.

23A.1 Document type master

Configurable under Administration. The type decides everything about how its documents behave.

Field	Type	Rule
Code / Name	nvarchar(20) / (80)	Unique.
Applies To	multi-select	Business Partner / Vehicle / Both. A type applying to partners can be narrowed to specific roles — Driving Licence applies only to the Driver role.

Field	Type	Rule
Is Expirable	bit	Drives whether Expiry Date is mandatory.
Default Validity	int + enum	e.g. 12 Months. Pre-computes the new expiry date on renewal.
Is Periodic	bit	True = the slot is expected to be renewed. Missing renewals are reported.
Renewal Lead Days	int	Default 30. Feeds notifications.
Mandatory For Activation	bit	Blocks vehicle activation (BR-VH-015) or warns on partner save (BR-BP-005).
Requires Document Number	bit	Insurance and permits yes; a cheque copy no.
Has Cost	bit	True = a renewal may post an expense and may link to a recurring charge (§19A).
Allowed Formats	list	Default PDF, JPG, PNG.
Max File Size (MB)	int	Default 10.
Retention Years	int	How long superseded versions are kept. Default 7.

23A.2 Seeded types and their periodicity

Document type	Applies to	Periodic	Typical validity	Mandatory
Registration Book	Vehicle	No	Life of vehicle	Yes, for Self Owned and Bank Leased
Insurance Policy	Vehicle	Yes	12 months	Warn
Fitness Certificate	Vehicle	Yes	6 or 12 months	Warn
Route Permit	Vehicle	Yes	12 months	Warn
Token Tax Receipt	Vehicle	Yes	12 months	No
Lease / Finance Agreement	Vehicle	No	Agreement term	Yes, for Bank Leased

Document type	Applies to	Periodic	Typical validity	Mandatory
Purchase Invoice	Vehicle	No	—	No
Driving Licence	Partner, Driver role	Yes	5 years	Yes
CNIC	Partner, Person	Yes	10 years or lifetime	Yes
NTN Certificate	Partner, Company	No	—	Warn
Service / Rate Agreement	Partner	Yes	Contract term	No
Cheque Copy	Partner	No	—	No

23A.3 Versioning and the renewal cycle

A file is never replaced in place. Uploading a renewal creates version $n+1$, marks it Current, and sets the previous version to Superseded with its own dates intact.

```
stateDiagram-v2
    [*] --> Active: Upload
    Active --> ExpiringSoon: Within lead days
    ExpiringSoon --> Expired: Expiry date passed
    ExpiringSoon --> Superseded: Renewal uploaded
    Expired --> Superseded: Renewal uploaded
    Active --> Superseded: Early renewal
    Superseded --> [*]: Retention period ends
```

ID	Rule
BR-DOC-001	Exactly one version per owner and document type carries <code>IsCurrent = 1</code> . Uploading a renewal flips the flag inside the same transaction.
BR-DOC-002	Superseded versions are never deleted by any user role. They age out only by the type's retention period, through a system job that logs what it removed.

ID	Rule
BR-DOC-003	Document status is recalculated nightly against the current date, so Active, Expiring Soon and Expired are always truthful without anyone opening the record.
BR-DOC-004	A vehicle with an expired document flagged <i>Mandatory For Activation</i> is flagged on the vehicle list and dashboard. Whether this blocks trip assignment is a configuration switch, defaulted to warn (OQ-15).
BR-DOC-005	The Renew action pre-fills type, number, provider and computes the new expiry from the previous expiry plus the type's default validity. The user supplies the new file and confirms the dates.
BR-DOC-006	When the type has Has Cost , the renewal screen offers to post the premium or fee as a vehicle expense, and to link it to the matching recurring charge entry (§19A) so the payment is not recorded twice.
BR-DOC-007	Every upload, download, renewal and retention-deletion is audited with user, timestamp and document version.
BR-DOC-008	A document may be marked Rejected with a reason — a scan too blurred to read, the wrong vehicle's papers. A Rejected version never becomes Current and does not satisfy a mandatory check.

23A.4 Screens

Screen	Contents
Partner / Vehicle → Documents tab	Current documents by type, with status chip, expiry date and days remaining. A Show history toggle reveals superseded versions per slot. Actions: Upload, Renew, Replace file, Reject, Download, Preview
Document Register	Fleet-wide and partner-wide list of every current document. Filters: owner type, document type, status, expiring within N days, branch, vehicle category, role. This is the compliance officer's screen
Missing Documents report	Every vehicle and partner lacking a document its type master marks mandatory or periodic — the gap list, which an expiry report alone cannot show
Expiry Calendar	Month view of upcoming expiries across the fleet, for planning renewals and their cash outflow
Bulk upload	Drag several files, tag each with owner and type in a grid, save once. Intended for go-live data loading

FR-DOC-001 — Preview renders PDF and images in-app. Downloads are served through a short-lived authorised URL; the storage path is never exposed.

23A.5 Storage and access

Area	Requirement
Storage	Files outside the database, in dated folders by owner type and id. Stored under a generated name; the original filename is metadata
Integrity	SHA-256 hash stored per version, so a file can be shown to be unaltered
Scanning	Virus-scanned on upload; a failed scan rejects the file and notifies Admin

Area	Requirement
Access	Download permission is separate from view permission. Driver CNIC and licence scans are restricted to Admin and Fleet Manager (OQ-17)
Audit	Every download records user, document, version, timestamp and IP
Encryption	Encrypted at rest; transport over TLS only

OQ-15 — Should an expired mandatory vehicle document hard-block trip assignment, or only warn?

OQ-16 — Confirm the retention period for superseded documents. Seven years is assumed.

OQ-17 — Who may download partner identity documents — Admin only, or Admin plus Fleet Manager?

23B. Role-based access control

Roles are data, not code. An administrator creates roles and ticks permissions; a user is assigned one or more roles and sees exactly the menus, screens, actions and amounts those roles permit. The matrices in §13.3 and §23.3 are shipped defaults, not fixed definitions — every line in them is a permission that can be moved between roles.

Access is controlled at three levels, because screen-level control alone is not enough here. A Fleet Manager needs the vehicle screen but must not see the purchase price on it.

Level	Controls	Example
Interface	Whether a menu, screen or tab is reachable at all	The Finance menu is hidden entirely
Operation	Whether an action can be performed on a screen the user can see	The vehicle screen opens, but Activate and Change Category are absent
Field	Whether a specific value is returned and displayed	The vehicle screen opens without purchase price, profit or driver salary

23B.1 Model

```

flowchart LR
    U[User] --> UR[UserRole]
    UR --> R[Role]
    R --> RP[RolePermission]
    RP --> P[Permission catalogue]
    UR --> SCOPE[Data scope: branch]

```

Entity	Purpose
Permission	Seeded catalogue of every gated capability. Code, name, group, level (Interface / Operation / Field). Not user-editable — the application ships the list
Role	Admin-defined. Name, description, is system role, is active
RolePermission	Which permissions a role grants
User	Login, linked Business Partner where the user is also a driver, status
UserRole	A user may hold several roles; effective permission is the union. Carries the data scope

23B.2 Permission catalogue

Business Partner

Code	Gates
BP.VIEW	Partner list and screen
BP.CREATE	Create a partner
BP.EDIT	Edit general and role details
BP.ROLE.MANAGE	Add or remove roles on a partner
BP.STATUS.CHANGE	Deactivate, reactivate, blacklist
BP.MERGE	Merge duplicate partners

Code	Gates
BP.DELETE	Soft-delete an unused partner
BP.FIELD.SALARY.VIEW	Driver salary, commission basis and value
BP.FIELD.CREDIT.VIEW	Credit limit, credit days, payment terms
BP.FIELD.OPENING.VIEW	Opening balance and its date
BP.EXPORT	Export the partner list

Vehicle

Code	Gates
VEH.VIEW	Vehicle list and vehicle screen
VEH.CREATE	Create and save a Draft vehicle
VEH.EDIT	Edit identity, technical and operational fields
VEH.ACQUISITION.EDIT	Step 3 — purchase price, amount paid, finance block
VEH.ACTIVATE	Activate a Draft vehicle, writing the opening postings
VEH.CATEGORY.CHANGE	Close a relation and open a new one
VEH.STATUS.CHANGE	Maintenance, temporarily unavailable
VEH.DISPOSE	Retire, sell, transfer
VEH.ITEM.MANAGE	Attach, detach and transfer attached items
VEH.DRIVER.ASSIGN	Set or release the default driver
VEH.FIELD.COST.VIEW	Purchase price, amount paid, acquisition cost, item cost
VEH.FIELD.FINANCE.VIEW	Finance amount, installment schedule, outstanding
VEH.FIELD.PROFIT.VIEW	Vehicle profit and loss, margin, income totals
VEH.EXPORT	Export the vehicle list

Financial operations

Code	Gates
FIN.EXPENSE.CREATE	Add a major or operating expense
FIN.EXPENSE.EDIT	Edit an unapproved expense
FIN.EXPENSE.APPROVE	Approve expenses above the configured threshold
FIN.INCOME.CREATE	Add vehicle income
FIN.INCOME.EDIT	Edit unposted income
FIN.INSTALLMENT.PAY	Record a lease installment payment
FIN.AGREEMENT.MANAGE	Create, amend, close or settle a finance agreement
FIN.RECURRING.MANAGE	Configure recurring charges (§19A)
FIN.RECURRING.AUTOPOST	Set a charge to Auto-post — Admin-only by default (BR-VH-028)
FIN.DUE.CONFIRM	Confirm a Due entry, posting its transaction
FIN.DUE.WAIVE	Waive a due entry with reason
FIN.ADJUSTMENT.POST	Post a correction or reversal (source §13)
FIN.PNL.VIEW	Trip, monthly and vehicle-to-date profit and loss
FIN.REPORT.VIEW	Financial reports and exports

Documents

Code	Gates
DOC.VIEW	See document lists and metadata
DOC.UPLOAD	Upload a new document
DOC.RENEW	Upload a renewal, superseding the current version
DOC.DOWNLOAD	Download or preview a file
DOC.DOWNLOAD.SENSITIVE	Download identity documents — CNIC, licence (OQ-17)
DOC.REJECT	Mark a version rejected

Code	Gates
DOC.REGISTER.VIEW	The fleet-wide document register and missing-documents report

Administration

Code	Gates
ADM.ROLE.MANAGE	Create roles and assign permissions
ADM.USER.MANAGE	Create users, assign roles and scope
ADM.MASTER.MANAGE	Lookup masters — vehicle types, expense types, document types
ADM.NOTIFICATION.MANAGE	Notification rules and lead days
ADM.SERIES.MANAGE	Numbering series
ADM.AUDIT.VIEW	Audit log viewer
ADM.CONFIG.MANAGE	System configuration switches

23B.3 Field-level rules

ID	Rule
BR-SEC-001	A field the user cannot see is excluded server-side , not hidden by CSS. The API response does not contain the value.
BR-SEC-002	Derived values that would reveal a hidden field are hidden with it. Without <code>VEH.FIELD.COST.VIEW</code> , a user sees neither the purchase price nor any total, chart or export that includes it. Without <code>VEH.FIELD.PROFIT.VIEW</code> , income may be visible but profit is not.
BR-SEC-003	Exports, printouts and API responses honour exactly the same field permissions as the screen. There is no export route around a hidden amount.

ID	Rule
BR-SEC-004	Hidden amounts render as a dash, not as zero, so no one mistakes a permission for a value.

23B.4 Data scope

A permission says *what* a user may do; scope says *on which records*.

Scope	Effect
All branches	No filter
Own branch	Lists, pickers, reports and totals filtered to the user's branch
Own vehicles	Only vehicles where the user is the assigned driver or nominated owner — used for the driver app
Own records	Only rows the user created — used for data-entry roles under review

Scope is set per user-role assignment, so the same Finance role can be all-branch for the head office user and single-branch for a regional one.

23B.5 Enforcement

ID	Rule
BR-SEC-005	Default deny. A capability with no permission granted is unavailable, and a new permission added in a future release is off for every role until an administrator grants it.
BR-SEC-006	Every API endpoint declares its required permission and checks it. The UI hiding a button is a convenience, never the control.
BR-SEC-007	A denied request is logged with user, permission, endpoint, record and timestamp, and repeated denials are reportable.

ID	Rule
BR-SEC-008	The built-in Administrator role cannot be deleted or stripped of <code>ADM.ROLE.MANAGE</code> , and the system refuses to leave zero active administrators.
BR-SEC-009	Permission changes take effect on the user's next request. Active sessions are re-evaluated rather than waiting for logout.
BR-SEC-010	Every role creation, permission grant or revoke, and user-role assignment is written to the audit log with before and after values.
BR-SEC-011	Separation of duties: <code>FIN.ADJUSTMENT.POST</code> and <code>FIN.EXPENSE.APPROVE</code> cannot be exercised by the user who created the record being adjusted or approved, when the corresponding configuration switch is on (OQ-19).

23B.6 Screens

Screen	Contents
Administration → Roles	Role list with user counts. Create, clone an existing role, deactivate
Role editor	Permission tree grouped by module, then by level — Interface, Operation, Field — with select-all per group and a search box
Administration → Users	User list, role assignment, branch scope, link to a Driver partner, activate and deactivate
Effective permissions viewer	For a chosen user, the flattened union of all their roles with the role that granted each permission. This is the screen that answers "why can he see that?"
Permission change log	Audit view of who changed which role's permissions and when

FR-SEC-001 — Cloning a role is the intended way to create a variant, since most new roles are an existing one plus or minus two capabilities.

23B.7 Default role templates

Shipped, fully editable. Each is a starting point for the client's own roles.

Role	Granted
Administrator	Everything
Fleet Manager	BP view/create/edit/role-manage, VEH all except acquisition-edit and dispose, VEH.FIELD.FINANCE.VIEW, DOC upload/renew/view, no cost or profit fields, no financial posting
Finance User	BP view + all field permissions, VEH view + acquisition-edit + all field permissions, all FIN permissions, DOC view/upload
Operations User	BP view, BP create limited to Customer roles, VEH view, VEH.DRIVER.ASSIGN, FIN.INCOME.CREATE, DOC upload, no cost, finance or profit fields
Driver	Own-vehicle scope, own profile, trip and fuel entry, no financial or partner access
Read Only	View permissions and exports across BP, VEH and DOC; no field permissions for cost, profit, salary or credit

OQ-18 — Should permissions ever be granted directly to a user, or only through roles? Roles-only is assumed, and is simpler to audit.

OQ-19 — Is maker-checker required on expenses and adjustments, and above what amount?

OQ-20 — Do driver app users sit in this same role system, or in a separate driver-only scheme?

24. Numbering, master data and non-functional requirements

24.1 Numbering series

Entity	Format	Reset	Example
Business Partner	BP-{YY}-{00000}	Yearly	BP-26-00147
Vehicle	VH-{YY}-{0000}	Yearly	VH-26-0032
Vehicle Transaction	VT-{YY}-{000000}	Yearly	VT-26-001204
Finance Agreement	FA-{YY}-{0000}	Yearly	FA-26-0011
Document	DOC-{YY}-{000000}	Yearly	DOC-26-000583

Prefix, padding and reset period are held in a series master so the client can change them without a deployment. Numbers are allocated inside the save transaction and are gap-free.

24.2 Configurable master data to seed

Lookups the client must populate before go-live. Each is maintained under Administration with code, description, sort order and active flag.

Master	Seed values needed
Vehicle Type	Truck, Trailer, Prime Mover, Tanker, Pickup, Bus, Car, Bike, Other
Make	Hino, Isuzu, Nissan, Mercedes, Master, Toyota, Suzuki, FAW, Shehzore, Other
Body Type	Open, Container, Tanker, Flatbed, Refrigerated, Dumper
Axle Configuration	4x2, 6x2, 6x4, 10-wheeler, 22-wheeler
Attached Item Type	Container, AC Unit, Tyre Set, Crane, Tanker Body, Refrigeration Unit, Tail Lift, Tracker Device, Other
BP Document Type	CNIC, Driving Licence, NTN Certificate, Agreement, Cheque Copy, Other — each with expirable flag

Master	Seed values needed
Vehicle Document Type	Registration Book, Insurance, Fitness, Route Permit, Token Tax, Lease Agreement, Purchase Invoice — each with expirable flag
Expense Type	Engine, Tyres, Body Work, Container, AC, Major Repair, Registration, Insurance, Fuel, Toll, Parking, Driver Allowance, Other
Income Type	Monthly hire, Trip income, Rental income, Other
Finance Type	Fully Paid, Bank Lease, Ijarah, Loan
City	Pakistani cities with province mapping
Branch	The client's own locations

24.3 Non-functional requirements

Area	Requirement
Performance	Partner and vehicle list screens return within 2 seconds at 10,000 partners and 2,000 vehicles. Vehicle save with a 60-row schedule completes within 3 seconds.
Concurrency	Optimistic concurrency on vehicle and partner edits; a stale save shows what changed and by whom rather than overwriting.
Transactionality	Vehicle activation is atomic across all eight tables. Any failure rolls back completely and leaves no partial vehicle.
Files	Stored outside the database with a path reference, virus-scanned, served through an authorised endpoint, never a public URL.
Data retention	No hard deletes of financial or lifecycle data, per source §13.

Area	Requirement
Localisation	English UI in V1. All labels and messages from resource files so Urdu can be added without code changes.
Time	Every timestamp is captured from the clock of the machine the user is working on, converted to UTC for storage, and rendered back in the time zone, calendar and date format of whichever machine displays it. The system has no fixed display time zone.
Backup	Daily full backup, point-in-time recovery to 15 minutes.

24.4 Date and time handling

ID	Rule
NFR-DT-01	System-stamped instants — created on, modified on, status changed on, audit timestamp — are taken from the user's machine clock at the moment of the action, converted to UTC and stored as <code>datetime2</code> in UTC.
NFR-DT-02	Every instant is rendered in the viewer's own time zone, calendar and locale format, resolved at render time from the browser or device. Two users in different zones see the same event at their own local time.
NFR-DT-03	Business dates the user types — acquisition date, transaction date, installment due date, document expiry, licence expiry, start and end dates — are stored as <code>date</code> , with no time and no time-zone conversion. A due date of 10-Oct-2026 stays 10-Oct-2026 everywhere.

ID	Rule
NFR-DT-04	Date pickers default to the user's machine date. "Not future" validations compare against the user's local date, not the server date, so a user ahead of UTC is not blocked from entering today.
NFR-DT-05	The API exchanges instants in ISO 8601 with offset (2026-10-10T14:30:00+05:00) and date-only values as plain YYYY-MM-DD . Formatting for display is done by the client, never by the server.
NFR-DT-06	Notification lead times are evaluated against the business date in the company's configured operating time zone, so a 3-day installment reminder fires on the same day for every user regardless of where they log in.
NFR-DT-07	Exports and printed documents carry the date in the exporting user's format, with the time zone named beside any timestamp.

Open question OQ-11 — NFR-DT-06 needs one operating time zone for the notification batch. Confirm this is Asia/Karachi.

25. Acceptance criteria

Each criterion is a UAT script. A module is accepted when every criterion passes.

Business Partner

ID	Given	When	Then
AC-BP-001	A new partner form	Party Type = Company is chosen	NTN is mandatory, CNIC is hidden
AC-BP-002	A partner with roles Workshop and Vendor	The vendor picker on an expense form is opened	The partner appears once, not twice

ID	Given	When	Then
AC-BP-003	An existing partner with CNIC 35202-1234567-1	A new partner is saved with the same CNIC	Save is blocked and the existing partner is offered
AC-BP-004	A partner named <i>Al-Madina Goods</i> in Lahore	A new partner <i>Al Madina Goods</i> in Lahore is entered	A similarity warning lists the existing record before save
AC-BP-005	A driver assigned to a vehicle	The Driver role is removed	Removal is blocked, listing the vehicle
AC-BP-006	A partner with one role	That role is unticked	VAL-BP-009 appears and save is blocked
AC-BP-007	A new partner with two contacts and a licence scan	Save is pressed once	All rows persist together and BP Code is shown
AC-BP-008	A driver licence expiring in 12 days	The notification run executes with a 30-day rule	An alert is raised for that driver
AC-BP-009	A Finance user	The partner screen is opened	Salary and credit limit are visible; a Fleet Manager sees neither
AC-BP-010	Legal Name changed from A to B	The History tab is opened	Old and new values, user and timestamp are shown

Vehicle

ID	Given	When	Then
AC-VH-001	A new vehicle wizard at step 1	Reg No LES-1234 is entered and LES 1234 exists	VAL-VH-002 blocks the save

ID	Given	When	Then
AC-VH-002	Category = Bank Leased	Step 3 is opened	The bank picker shows only Bank-role partners, and the finance block is mandatory
AC-VH-003	Purchase price 5,000,000 and paid 2,000,000	The vehicle is activated	The vehicle screen shows paid-to-date PKR 2,000,000, and two transactions exist
AC-VH-004	The vehicle from AC-VH-003	An installment of 150,000 is recorded	Paid-to-date immediately reads PKR 2,150,000 with no batch run
AC-VH-005	Installment 75,000, tenure 36, monthly, first due 10-Oct-2026	Step 3 is completed	36 schedule rows are previewed, last due 10-Sep-2029, total payable 2,700,000
AC-VH-006	A lease with first due on the 31st	The schedule is generated	February rows fall on the 28th or 29th, not rejected
AC-VH-007	Category = Self Owned, no registration book uploaded	Activate is pressed	Activation is blocked with VAL-VH-014
AC-VH-008	An active vehicle in category Rented	Category is changed to Self Owned on 01-Nov-2026	The rented relation closes 31-Oct-2026, a new relation opens, and prior rent transactions are unchanged
AC-VH-009	A vehicle saved as Draft with purchase data	The Draft is viewed	No transactions exist; they appear only after activation
AC-VH-010	A container costing 450,000 added at creation	The vehicle is activated	The attached item exists and a major expense of 450,000 is posted

ID	Given	When	Then
AC-VH-011	Driver Ali is default driver of LES-1234	Ali is set as default driver of LES-5678	VAL-VH-017 prompts to release the first assignment
AC-VH-012	A vehicle with 3 transactions	Delete is attempted	Delete is unavailable; only retire, sell or transfer are offered
AC-VH-013	An Operations user	The vehicle screen is opened	Purchase price and finance figures are hidden
AC-VH-014	Insurance expiring in 25 days with a 30-day rule	The notification run executes	An alert is raised naming the vehicle and document

26. Assumptions, decisions and open questions

Decisions taken in this FSD

These were not specified in the source document. They are written as decisions so development is not blocked; each can be reversed cheaply if the client disagrees.

#	Decision	Why
D-01	Partner form is tabbed; vehicle form is a five-step wizard	Partners are edited often; vehicles are inducted rarely and carry financial consequences
D-02	Initial payment is posted as a transaction, not stored on the vehicle	The only way the §8 example works without a manual total
D-03	Down payment must equal amount paid at creation	Prevents double counting of the same money
D-04	Draft status for vehicles	Purchases in progress need a home before the papers arrive

#	Decision	Why
D-05	Bank, Workshop, Body Maker, Tracker and Fuel Card roles get no dedicated detail table in V1	Their attributes fit the common fields; adding tables later is additive
D-06	Category change closes and supersedes the relation	Source §3 requires history never to be overwritten
D-07	Opening odometer captured at creation	Makes every later fuel-efficiency and distance figure meaningful

Open questions for the client

ID	Question	Blocks
OQ-01	Are vehicle numbers ever re-registered to a new number, and must both numbers stay searchable?	BR-VH-017
OQ-02	Do partners go live with opening balances, or does everyone start at zero?	Field 17, §6
OQ-03	Do Bank, Workshop, Body Maker, Tracker or Fuel Card roles need fields beyond those in §7.2–7.7?	D-05
OQ-04	Should a missing driving-licence scan block saving a driver, or only warn?	BR-BP-005
OQ-05	For Shared vehicles, are expenses split at entry time or only at month-end P&L?	§17.1 expense-sharing rule
OQ-06	Is partner merge needed in V1, or can it wait for Phase 1.1?	FR-BP-017
OQ-07	Should lease installments split principal and markup, or stay a single amount in V1?	§18.2
OQ-08	On a Bank Leased vehicle, is the down payment treated as an asset cost or a finance payment for P&L?	§19, source §10
OQ-09	Are there existing partner and vehicle records to migrate, and in what format?	Data migration scope

ID	Question	Blocks
OQ-10	Which branches exist at go-live, and should users be restricted to their own branch by default?	§6 field 15, visibility

Assumptions

- Single company, multiple branches. Multi-company is not required in V1.
- PKR only. No multi-currency.
- No accounting integration in V1; the vehicle transaction ledger is operational, not a general ledger (source §10 makes the same distinction).
- Web application for admin users, separate mobile app for drivers, sharing one API.
- Users authenticate with username and password; SSO is not required.

Next steps

1. Client reviews this document and answers OQ-01 to OQ-10.
2. Screen wireframes for the partner tabs and the five wizard steps.
3. Database schema script and EF migrations for the seventeen tables named here.
4. Build order: master data → Business Partner → Vehicle → vehicle transactions → notifications.